

Osaid Khan

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Dear Hiring Team,

It is with great pleasure that I submit my candidacy for the **Software Engineer** position with **Applied Materials**. I bring over three years of professional experience in high-performance backend systems, systems-level engineering, and multi-threaded orchestration, along with a strong academic foundation as a Master of Science in Computer Science graduate from Pace University, NY.

Through my experience across two engineering roles, I have built and delivered production-grade software that directly reflects **the rigorous engineering of high-performance software modules and libraries for semiconductor equipment**. As a Software Engineer at Sperse, I architected a **multi-threaded** orchestration system using **LangChain** and **LangGraph** to enable reliable coordination across complex industrial workflows — requiring the same focus on system quality and functionality that Applied Materials demands. To ensure standardized implementation across products, I designed **reusable software modules** and **SDK libraries** that accelerated system integration while maintaining rigorous **software documentation** and **test procedures**. Prior to Sperse, as a Software Engineer at Qualitest, I developed a **high-availability Storage Service** using **Python** and **Redis** that implemented fundamental **networking principles** for distributed operations. I also engineered **common software modules** using **AWS Step Functions** to manage long-running jobs, preventing system timeouts through robust **concurrency management** — both of which required **the kind of systems-level thinking and operational stability focus this role demands**. Together, these roles have prepared me to **design, prototype, and develop complex software solutions while ensuring long-term system stability** within your engineering team.

My academic research experience has given me a rigorous foundation in GPU-accelerated computing, low-level optimization, and technical communication — skills that translate directly into **the focus on technical excellence, integrity, and the collaborative innovation Applied Materials is known for**. As an AI Researcher at the Pace Artificial Intelligence Lab, I leveraged **CUDA** and **PyTorch** on an **HPC cluster** to develop generative models, utilizing **low-level GPU programming** and **multi-threading** to solve moderately difficult technical problems. Beyond building the systems myself, I led technical workshops on **GPU programming**, **Mojo**, and **Algorithms** for an audience of 50+ students. Designing and delivering those sessions required me to translate complex requirements into clear technical concepts — directly mirroring **the technical leadership and customer-centric requirement analysis central to this role**. This experience has shaped me into an engineer who builds with rigor and communicates with clarity, both of which I would bring to **Applied Materials and its mission to lead in materials engineering solutions**.

I have always been passionate about **innovation and customer commitment** and envision myself contributing my talent to the furthering of **making possible the technology that transforms the future**. I would love to discuss my interest in this position with your team and can be reached at +1 (945) 304-5781 or khanosaid726@gmail.com. Thank you for your time and consideration, I look forward to connecting!

Sincerely,
Osaid Khan